

AE 460 Interim Report

Solar Gravitational Lens Project Proposal

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The following paper discusses the design parameters for a deep-space mission developed by Team Manifest Destiny at San Diego State University (SDSU). The purpose of this mission is to capture high-resolution images of exo-planets using Solar Gravitational Lens (SGL) technology, and perform spectral analysis on these images to test the possibility of life existing on these planets. This report will go over technical design parameters for the payload, data handling, electrical, Telemetry tracking and control, flight dynamics, ADCS, mechanical and thermal sub systems for this mission. Major design choices have been performed regarding each subsystem, including simulation. Moving forward, simulation for particular systems will be further studied such as propulsion systems and STK simulations to optimize travel time. A miniature model of the payload design will be constructed moving forward to prove concept design.

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I. System Overview

This paper will parameterize each system, going over details for Payload, Data Handling, Electrical, Flight Dynamics, Propulsions, ADCS, Mechanical, Thermal, and Telemetry Tracking and Control. They will each go over a simplified overview, then requirements, candidates and any simulations that could be involved.

Using the Sun's immense gravitational field as a natural lens, this mission seeks to assess the possible habitability of exoplanets through new estimations of their size, orbital parameters, atmospheric composition, and surface conditions. The primary payload of this mission will be the Solar Gravitational Lens (SGL). This is a meter-class telescope that is equipped with an internal coronagraph, spectrograph, and optical communication system that is designed to operate within the SGL focal region between 650 and 900 astronomical units from the Sun. Using Einstein's theory of General Relativity, the SGL will capture and analyze the light that is amplified and focused by the Sun's gravity to reconstruct

multi-pixel images with high resolution and spectra of distant exoplanets. The intent is that the SGL will reveal continents, oceans, and atmospheric bio-signatures. For this mission we will be looking at Proxima Centauri B (PCB), an exoplanet which is 4.24 light years from Earth. This planet was chosen due to its existence within the goldilocks range, and having close neighbors allows for other possible options within its system. The mission development timeline consists of over 20 years of research and development to mature critical future technologies such as deep-space communications, payload, and radiation-hardened electronics, followed by a 24-month build and integration phase involving material acquisition, testing, and spacecraft assembly; after launch, the spacecraft undergoes a 25-year autonomous travel phase with periodic system health checks, culminating in a data collection and processing phase lasting at least 10 years, resulting in a total estimated mission duration exceeding 55 years.

II. Payload Subsystem

The Payload of the SGL is one of the least field tested subsystems on board this mission, with extremely high levels of precision, mirrors with minimal to no damage throughout the journey, a coronagraph capable of high levels of light blocking, and plenty more. The Payload will undergo three stages throughout the journey; a stored stage during its transit to 650 AU, an operational stage for 13 years until it becomes less effective to operate, and a final shutdown phase where all data is transmitted and the payload will become nonoperational at about 900 AU.

A. Requirements

The Payload functions along a fairly simple pipeline. Light is reflected from a primary mirror into a secondary which concentrates the incoming light into a hole in the primary, similar to the method used on board the James Webb Space Telescope. Light from these mirrors is split using beam splitters between the science, TT&C, and ADC subsystems, since the light from earth is taken in as up link signals and will be sorted in its own section of the On Board Computer (OBC). The remaining light that is directed to the science will filter through a coronagraph to remove light from the sun disk. After, 3D Hyperspectral Imaging is then performed by collecting light across the beam and organizing it into a 2 spatial dimensions and one spectral, which creates a 'fingerprint' of the different objects in the image. Finally, an Image Acquisition step turns the light into digital signals which is then delivered to the OBC for image processing [1].

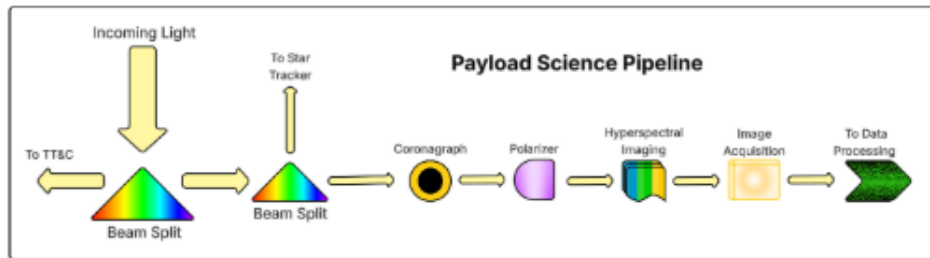


Fig. 1 Payload Science Pipeline, simplified

The primary and secondary mirrors of the Payload take up the majority of the weight for the subsystem. With extreme precision needed and the high forces of gravity utilized during the Sun Transfer leaving the Solar System, the mirrors will need to be supported and stabilized. The mirror will need to have a focal length of at least 1.2, and an effective focal length at the coronagraph of 10 m, in order to have an appropriately sized coronagraph that will be efficient at blocking light. The primary mirror will therefore need to be a minimum of 1.2 m from the secondary in order to focus at that distance. The mass of this will take the most of the allowed Payload size. A maximum mass of 50 kg will be allotted for both mirrors.

The Lyot coronagraph itself is expected to filter at a suppression of $1e-7$, which will account for the extreme brightness of the solar disk and corona while the light of the Einstein Ring will remain unaffected. This, with optics, a detector, and mechanisms to constrict the size of the blocking will lead to an expected mass of 15 kg. A Polarizer has been considered to be used in a few projects, however current projections suggest that the coronagraph will be efficient enough to leave the polarizer redundant and a bottleneck. Hyperspectral Imaging will require a spectral range of 0.5-1.5 μm , and a resolution of 50-150. The range represents wavelengths of light captured by the imaging, and covers visual light to beyond IR. The resolution is how thinly the light will be divided. These divisions will separate into a minimum of 50 channels for processing. Acquisition device is one that can obtain a high SNR, and would therefore be ideal for the

accuracy required in an SGL mission. This is why a Charge-Coupled Device (CCD) is ideal, which converts light into a highly accurate electrical pattern before digitizing the feedback [2]. After image acquisition, the data is moved to image processing.

B. Candidates

The mirror segments need to withstand considerable forces during the journey. Because of this, large amounts of mass in the system are taken up by stabilizers and support systems to keep these fragile devices safe. This was estimated to be done with 50 kg for a 1m mirror in 2013 [3], and as we are expected to launch in 20 years after further Research and Development, it is expected that we would be able to lower the weight cost of this system to at least 40 kg, saving room for other components of the payload. By carefully arraying the primary and secondary mirrors, we can save more mass per mirror. The material for this specific mirror system should be one that is light and effective with minimal wear over time. That is why the same paper developing the mirror suspension system also utilizes Zerodur (a glass ceramic) for its material. This choice also boasts very minimal thermal expansion, which is important for missions in deep space near absolute zero.

Following previously sourced coronagraph requirements [4], we are looking for $1e-7$ suppression of light at the Einstein ring. This is why most papers cite JPL's simulation for coronagraph testing [5] as the primary source of realistic modern and upcoming coronagraph tech. Future work in the field will require a more practical masking of the sun. However, this Lyot coronagraph is still extremely effective. For a 1 m aperture, this will boast $2e-7$ suppression, and for a 2 m aperture it can suppress up to $4.5e-8$ of the corona, which is far above expected figures. This is not currently a built device, but the technology currently exists for this required coronagraph. With the aforementioned focal lengths, there is an expected sun diameter for the coronagraph of 0.14 mm, which is a reasonable size for sun blocking, therefore the work proposed by Zhou [5] is acceptable for our mission requirements.

There are many current Hyperspectral Imagers that would be appropriate for the job. A reliable, field tested candidate would be the Pixxel spacecraft hyperspectral imaging [6]. This model has the potential to be adjusted to fit the required resolution and channels to properly extrapolate spectral data from incoming light. An ideal Image Acquisition device is one that can obtain a high SNR, and would therefore be ideal for the accuracy required in an SGL mission. This is why a Charge-Coupled Device (CCD) is ideal, which converts light into a highly accurate electrical pattern before digitizing the feedback [2]. Further analysis on the required tools for Data Processing and the OBC to be included in the OBC section.

C. Simulation

Simulation for the integration of Payload candidates is integrated as blur in the Image Processing, using calculations for SNR, blur, and error in the images as a result of imperfections and limitations to the technology. The current Payload simulation is explained in detail within the Data Handling Subsystem below. The method in which the values from the mirrors, coronagraph, etc. are integrated is a current in-progress task, however it is used to formulate the SNR of the deconvolution penalties.

$$\frac{SNR_R}{SNR_C} \approx \frac{\pi}{4 \ln(\sqrt{2} + 1)} \frac{D}{Nd} \approx 0.891 \frac{D}{Nd} \quad (1)$$

Much of the variables in this equation, are driven by the sampling strategies and errors introduced by the Image collection itself [7]. Results from the Payload Image processing is explained in the following section. However, it should be noted that further effort will be required to fully apply these conditions to the SNR and the above equation. This will be a primary task for the following months of effort.

III. Data Handling Subsystem

The data handling and onboard processing subsystem is responsible for acquiring, storing, and processing optical measurements collected at the focal region of the Solar Gravitational Lens (SGL). Unlike a conventional telescope, the SGL does not form a direct image of a distant exoplanet. Instead, light from the source is mapped into a highly distorted Einstein ring surrounding the Sun. Recovering a two-dimensional image therefore requires extensive post-processing and multiple spatial measurements accumulated over time.

To address this challenge, a software pipeline has been developed to simulate SGL image formation, tiled data acquisition, and image reconstruction. This pipeline serves both as a proof of concept for the image recovery process and

as a functional model for the onboard data handling and processing tasks expected during the mission. The equations in this section are available in [7].

A. Requirements

The data handling subsystem must support long-duration data acquisition, high-capacity onboard storage, and computationally intensive image reconstruction while operating under strict power and thermal constraints. The system must also be robust to communication delays and data dropouts, necessitating autonomous processing and verification prior to downlink.

1. Image Formation and Forward Model

Gravitational lensing by the Sun maps light from a distant source into a bright annular structure in the image plane. The relationship between the source-plane position and the image-plane position is governed by the gravitational lens equation:

$$\mathbf{y} = \mathbf{x} - \alpha(\mathbf{x}) \quad (2)$$

where $\mathbf{x}$ is the image-plane coordinate, $\mathbf{y}$ is the source-plane coordinate, and α is the deflection angle produced by the Sun's gravitational field.

A natural length scale for this system is the Einstein radius in the lens plane,

$$\xi_0 = \sqrt{\frac{4GM}{c^2} \frac{D_d D_{ds}}{D_s}}, \quad (3)$$

where G is the gravitational constant, M is the solar mass, c is the speed of light, D_d is the observer–lens distance, D_s is the observer–source distance, and D_{ds} is the lens–source distance. All coordinates are expressed in dimensionless form by scaling with ξ_0 ,

$$\mathbf{x} = \frac{\boldsymbol{\xi}}{\xi_0}, \quad \mathbf{y} = \frac{\boldsymbol{\eta}}{\eta_0}. \quad (4)$$

Because each point on the source contributes light around the full circumference of the Einstein ring, the image observed at the detector is a convolution of the source brightness distribution with the SGL point spread function (PSF). For a monochromatic source, the SGL PSF is given by

$$\text{PSF}(\rho) = \mu_0 \left[J_0^2 \left(\frac{2\pi\rho}{\lambda} \sqrt{\frac{2GM_\odot}{c^2 z}} \right) + J_1^2 \left(\frac{2\pi\rho}{\lambda} \sqrt{\frac{2GM_\odot}{c^2 z}} \right) \right], \quad (5)$$

where ρ is the radial coordinate in the image plane, λ is the wavelength, z is the heliocentric distance of the spacecraft, J_0 and J_1 are Bessel functions of the first kind, and μ_0 is the peak magnification factor.

2. Image Reconstruction and Deconvolution

The observed image is related to the true source image through convolution with the PSF. Recovering the source image therefore constitutes an inverse problem. In the Fourier domain, a regularized deconvolution may be written as

$$\hat{I}_{\text{src}}(\mathbf{k}) = \frac{\hat{H}^*(\mathbf{k})}{|\hat{H}(\mathbf{k})|^2 + \gamma} \hat{I}_{\text{obs}}(\mathbf{k}), \quad (6)$$

where $\hat{H}(\mathbf{k})$ is the Fourier transform of the PSF, γ is a regularization parameter, and $\hat{I}_{\text{obs}}$ and $\hat{I}_{\text{src}}$ are the Fourier transforms of the observed and reconstructed images, respectively. Regularization is required to suppress noise amplification in spatial frequency regions where the transfer function is small.

Successful image reconstruction depends critically on achieving sufficient signal-to-noise ratio (SNR) prior to deconvolution. In the background-dominated regime relevant to SGL observations, the SNR scales approximately as

$$\text{SNR} \approx \frac{N_s}{\sqrt{N_b}} \sqrt{T}, \quad (7)$$

where N_s is the signal photon rate, N_b is the background photon rate dominated by residual solar corona light, and T is the integration time. This relationship highlights that image recovery is fundamentally limited by photon statistics rather than computational speed, necessitating long integration times to enable stable inversion.

3. Implications for Onboard Processing

These physical constraints directly inform the design of the data handling subsystem. While deconvolution and reconstruction can be performed efficiently using parallelized onboard computation, the dominant mission timescale is set by photon collection. Consequently, the onboard processing system must support sustained data accumulation, intermediate storage of ring measurements, and autonomous reconstruction and verification prior to downlink. The simulation developed for this work demonstrates the feasibility of these processing tasks and provides a functional model for the expected onboard data handling pipeline.

B. Candidates

The data handling and onboard processing subsystem must support long-duration autonomous operation, high-throughput numerical processing, and power-aware execution while operating at extreme heliocentric distances. In evaluating candidate architectures, emphasis was placed on computational capability, energy efficiency, reliability, and compatibility with radiation-tolerant system design practices.

A hybrid architecture was selected as the most suitable approach. This architecture separates system supervision and low-level control from computationally intensive image processing tasks, allowing each function to be optimized independently.

1. Supervisory Control and Low-Level Data Acquisition

An Arduino-class microcontroller is selected to serve as the supervisory control element of the data handling subsystem. This device is responsible for managing sensor interfaces, coordinating data acquisition timing, monitoring system health, and enforcing power and thermal constraints. Its low power consumption (on the order of 1 W or less) and deterministic real-time behavior make it well suited for continuous operation over long mission durations.

The microcontroller maintains awareness of available electrical power by interfacing with the spacecraft power subsystem and monitoring current draw. It enforces a hard upper limit of approximately 25 W for the image processing subsystem while maintaining a nominal operating target of approximately 75% of the available power budget. If power availability decreases, the microcontroller issues commands to reduce computational load, delay nonessential processing, or temporarily suspend reconstruction while continuing raw data acquisition.

2. High-Performance Onboard Processing

Computationally intensive tasks are assigned to a Jetson-class single-board computer (SBC). This platform provides a Linux-based operating environment, multi-core CPU resources, and an integrated GPU capable of CUDA acceleration. These features enable efficient execution of data-parallel workloads such as point spread function evaluation, Fourier transforms, and regularized deconvolution.

GPU acceleration is used selectively to minimize power consumption while maximizing throughput. Tasks with high arithmetic intensity and minimal branching are executed on the GPU, while system coordination, scheduling, and adaptive logic remain on the CPU. This division allows the processing workload to scale dynamically within the enforced power envelope.

3. Adaptive and Multi-Resolution Reconstruction

To further manage computational load and energy consumption, the onboard processing system employs a multi-resolution reconstruction strategy. Initial image reconstruction is performed at low spatial resolution, emphasizing low-frequency content that converges rapidly and requires shorter integration times. These coarse reconstructions are used to verify signal presence, estimate convergence, and identify regions of interest.

From an algorithmic perspective, this approach corresponds to progressively extending the spatial frequency bandwidth of the reconstruction while adjusting the regularization parameter to maintain numerical stability. This strategy is particularly well suited to the ill-conditioned nature of the SGL inverse problem, where high-frequency components are the most sensitive to noise.

4. Storage and Data Management Considerations

Due to limited downlink bandwidth and long communication delays, significant onboard storage capacity is required. High-capacity solid-state storage, such as NVMe devices, provides sufficient throughput for ring data accumulation and

intermediate reconstruction products. Data are retained onboard until reconstructed images are verified and scheduled for transmission, after which raw measurements may be selectively discarded to free storage capacity.

Redundancy and data integrity are achieved through software-level verification and staged data retention rather than continuous mirroring, minimizing power overhead while preserving reliability.

5. Relationship to the Simulation Framework

The simulation developed for this work implements the same physical models and reconstruction algorithms as the proposed flight architecture but executes them in a batch-oriented, CPU-based environment. While the simulation validates mathematical correctness and feasibility, the operational architecture defines how these computations would be distributed over time, constrained by power availability, and executed autonomously.

Together, the simulation and candidate architecture demonstrate a realistic pathway from proof-of-concept image reconstruction to a fully autonomous, power-aware onboard data handling system.

C. Simulation

To evaluate the feasibility of the proposed data handling and image reconstruction pipeline, a numerical simulation was developed in C++. The simulation is intended to validate the physical modeling, data flow, and reconstruction methodology of the Solar Gravitational Lens (SGL) imaging process rather than to serve as a full mission-level performance model.

A high-resolution image of Earth was used as a proxy for a distant exoplanet and treated as the ground-truth source image. This image represents the idealized spatial information that the SGL system seeks to recover. The simulation then applies a simplified forward model to map this source image into a set of Einstein ring measurements consistent with gravitational lensing and diffraction effects.

The simulation proceeds through the following stages:

- 1) The source image is subdivided into a grid of tiles, representing discrete spatial samples of the object.
- 2) Each tile is mapped into an annular contribution to an Einstein ring, reflecting the azimuthal encoding inherent in SGL image formation.
- 3) A radial point spread function is applied to model diffraction effects at the SGL focal region.
- 4) Ring measurements are accumulated and stored, emulating long-duration photon integration.
- 5) A regularized deconvolution algorithm is applied to reconstruct a two-dimensional image from the simulated ring data.

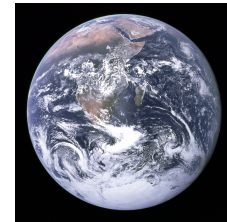


Fig. 2

The simulation is parallelized using OpenMP and executes on CPU hardware to enable rapid testing and validation. While the operational system would employ GPU acceleration and streaming reconstruction, the simulation performs reconstruction as a batch process after all measurements have been generated. This approach simplifies validation while preserving the underlying physics and reconstruction logic.

Importantly, the simulation does not attempt to model all sources of noise, spacecraft motion, or hardware imperfections. Instead, it focuses on demonstrating that the inverse reconstruction problem is computationally tractable and that meaningful spatial information can be recovered from annular SGL measurements. As such, the simulation serves as a proof of concept for the full onboard data handling and image processing pipeline described in this work.

D. Results

The results of the simulation demonstrate the feasibility of reconstructing a two-dimensional image from Solar Gravitational Lens (SGL) measurements using the proposed data handling and image processing pipeline.



(a) Ring constructed from a section of the earth image. (b) Ring constructed from a different section of the earth image (c) Reconstructed Earth Image

Representative Einstein ring measurements generated by the simulation are shown in Figures 3a and 3b. These images illustrate the highly distorted nature of raw SGL observations, in which spatial information from the source is encoded azimuthally around the ring. Individual ring measurements do not resemble the original object and cannot be interpreted directly without reconstruction.

Figure 3c shows the reconstructed image produced by aggregating and processing the simulated ring measurements. Despite the severe distortion present in the individual rings, the reconstruction successfully recovers large-scale spatial features of the original source image. Continental boundaries and major brightness variations are preserved, demonstrating that meaningful spatial information can be recovered through regularized deconvolution.

These results confirm that the SGL image reconstruction problem is computationally tractable and that onboard processing resources are sufficient to perform reconstruction autonomously. While the simulation does not include all sources of noise, spacecraft motion, or hardware-specific effects, it validates the core reconstruction methodology and illustrates the expected behavior of the system under idealized conditions.

The simulation and reconstruction pipeline described in this work are implemented in an openly available codebase hosted on GitHub [8]. The repository contains both the simplified CPU-based simulation used to generate the results presented here and a more complete hybrid codebase intended to represent the operational onboard architecture. The latter includes power-aware scheduling, adaptive multi-resolution reconstruction logic, and a supervisory control framework reflecting the interaction between a microcontroller-based control layer and a GPU-accelerated processing system.

In an operational mission, this full pipeline would execute incrementally as measurements are accumulated over time, producing low-resolution reconstructions early and progressively refining image detail as signal-to-noise ratio improves. The results presented here therefore serve as a proof of concept for the complete data handling subsystem and demonstrate the feasibility of autonomous onboard image reconstruction for SGL missions.

IV. Electrical Subsystem

The EPS must manage the energy storage, power distribution, and conditioning. For a long deep space mission (30+ year duration) beyond the solar system, the functionality of solar panels becomes exponentially inefficient as the space-craft strays from the sun. Due to this circumstance added with the longevity of the mission, EPS must rely on nuclear power as its main source of power, such as Radioisotope Power Systems (RPS) [9]. Although nuclear power within a limited weight requirement will produce a minor amount of power. Combined, the systems will be constrained to a peak load of 78 Watts, with power switching from Propulsions to OBC after the flight path is managed. It also provides waste heat, which can be harnessed to regulate the temperatures of the payload systems [10].

A. Requirements

The EPS shall provide continuous regulated electrical power to all spacecraft subsystems for the full mission duration exceeding 35+ years. The EPS must support a maximum spacecraft electrical load of 78 W while maintaining a regulated 28 V DC power bus. Power availability shall be continuous, with prioritization logic implemented such that

all subsystems are properly maintained under various operational modes. Certain systems will undergo load shedding, such as propulsion and power, which will be continuously regulated to other systems.

EPS will be resistant to single point failure via protected switching and modular power distribution. The system shall operate autonomously with regulation via the OBC. Radiation-hardened electronics shall be used to ensure survivability and longevity.

B. Candidates

Table 1 Spacecraft Power Requirements by Subsystem

Subsystem	Power Requirement (W)
Telemetry System	15
Payload	10
ADCS	28
Propulsion System	20
Onboard Computer	25

The power requirements are deterministic of the use of single system technology for each section. Propulsion wattage requirements will be at maximum 20 watts during the flight-path, which will level off and switch into the use of the OBC. When looking into voltage historically, long-duration probes (e.g., Voyager and Pioneer) have used 28 V DC as the optimum regulatory voltage throughout the bus [11]. DC converters and mechanical switching systems and shunt regulators will be used to control bus voltage.

The spacecraft will use a modular storage design comprised of a Nickel-Hydrogen (NiH_2) cell system used to buffer transient loads and provide stability to power during load switching events. Conservative depth-of-discharge (DOD) and cycle times are chosen via NASA design standards. In this case, to reach 15,000+ cycles, the battery will be limited to 40 percent DOD maximum [12]. This will include a built-in power conditioning unit (PCU) to provide control electronics for the EPS. Shunt and DC regulators previously mentioned will be implemented to control output voltage and load isolation. Protective elements such as fuses and relays will be installed.

The main generator will consist of a Multi-Hundred Watt Radioisotope Thermoelectric Generator (MHW-RTG). This generator harnesses the thermal radiation of Pu-238 converted into electric energy via thermocouples.

This particular model was used during the Voyager 1 mission, demonstrating over 38 years of continuous operation. This generator is designed to have a mass of 37.7 kg with compact dimensions, while producing approximately 157 W of electrical power at beginning of life [13]. Power output degrades over time due to the half-life of plutonium and thermoelectric material degradation, reducing output by approximately 1.6 percent per year. With a Pu-238 half-life of 43.32 years, the generator is capable of supplying the required 78 W throughout the mission duration.

The MHW-RTG power shall be directed through a Nickel-Hydrogen battery cell configuration, which will distribute regulated power to all spacecraft subsystems via optimized wiring and power management control.

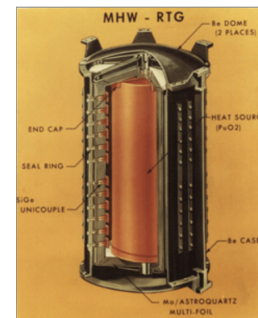
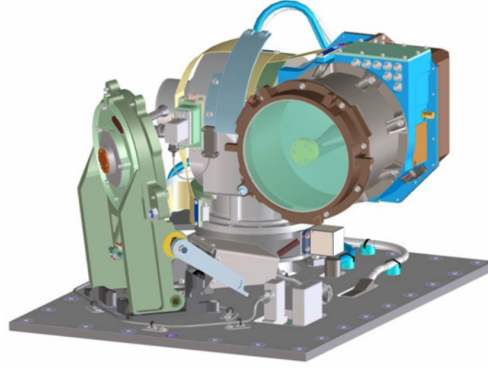


Fig. 4 NASA's MHW-RTG Design

V. Telemetry, Tracking, and Control Subsystem

The spacecraft will be operating a cutting-edge optical communication system, which will use lasers to send packets of data to the ground stations. These ground stations will uplink using powerful 1550 nm lasers to punch through Earth's atmosphere with an accuracy within a few tens of nanoradians and at a high spectral purity. The spacecraft's 1-m payload telescope will receive the signal. The downlink will be sent by the spacecraft's 15 W laser outputting a 1064 nm frequency towards earth. The downlink transmission supports an 8 kbps data rate. The difference in uplink and downlink frequencies will minimize the impact of solar radiation on the laser communications.



R0.35

Fig. 5 Inertially Stabilized Optical Module

A. Requirements

Supporting a science data rate of 16 Gbits/year from a 900 AU distance is the primary goal of the communications system. Maximizing line of sight and transmission to support this data rate has helped frame the requirements for the ground station system design. Requirements include that the communication system shall not exceed a 20 Watt power draw, it shall not exceed 10 kg, and it shall have a communications link margin greater than 4.

B. Candidates

1. Ground Stations

An analysis that compared 5, 10, and 30 meter class ground stations was performed for a case at 900 AU[1]. Based on these results, a 30-m class optical ground station is necessary to support the needed data rates as long as the spacecraft is transmitting for 30 percent of the year from 900 AU. Three ground stations around the Earth will be positioned such that a remote spacecraft near the ecliptic plane can always see one of the ground stations as long as Earth isn't blocked by the sun. Goldstone (USA), Canberra (Australia), Robledo (Madrid, Spain) are all sites being used by NASA's Deep Space Network, and an STK simulation has shown they can support a 24/7 line of sight. Building 30-m ground stations at these sites would be convenient for ground station personnel and would support the infrastructure for future deep-space optical communication systems. During end-of-life operations, these ground stations will be repurposed to support optical communications in other deep space missions.

2. Hardware and Software

RF communications are unusably weak at 950 AU and would exceed the mass budget. The current state of the art for optical communications hardware is represented by NASA's Laser Communications Relay Demonstration Project (LCRD). The SGL's communication system will be modeled after this technology, developed, and then customized for the SGL Spacecraft. The modulation, coding, and detection scheme will be Photon Counting and Pulse Position Modulation (PPM), a proton-efficient design optimal for deep space missions. The uplink communications signaling is 4-ary PPM with a two comparator demodulator performing binary hard decisions. The gimbal-stabilized optical module is pictured here.

The National Aeronautics and Space Administration (NASA) at Glenn Research Center (GRC) has developed an optical communications transmitter and receiver using Commercial-off-the-shelf (COTS) components. This project is known as The Real Time Optical Receiver (RealTOR) project[14]. The PPM flight transmitter will encode data using a rate- $\frac{1}{2}$ serially concatenated PPM turbo code, this data is convolutionally interleaved, and then modulated with a 16-ary PPM modulation scheme. A master-oscillator power amplifier (MOPA) architecture will be used in which modulation will be performed using a Mach-Zehnder modulator and amplified with a two-stage Erbium-doped fiber amplifier (EDFA). The PPM flight receiver is an optically pre-amplified direct detection receiver. Received uplink is de-interleaved and decoded.

3. Setup and Operation

Four atmospheric sensors will continually support the ground station during nominal operations. A weather station, sunphotometer, ground scintillometer, and a cloud imager will work as an integrated suite to characterize the optical channel[15]. High-capacity connections will allow all three ground sites to interact 24/7. A Mission Operations Center at the custom ground stations will perform planning and scheduling, control, status monitoring, and reporting and accountability services. Telemetry and ground stations status' will be tracked and relayed to controllers, engineers, and scientists using an Epoch suite of telemetry and commanding software. At least 2 controllers will be on staff at each ground site during nominal operations, along with a spacecraft engineer and ground engineer on call. A total staff size of 25 people per ground site throughout the year. Nominal operations will occur 24/7 until the end of the mission. A downlink will be received for 30 percent of the year during science data transmission periods, which happen when the exoplanet is in view. The spacecraft will be almost fully autonomous, but will periodically update telemetry through downlink and receive uplink from Earth. The main purpose of uplink commanding will be to make sure the spacecraft's clocks are aligned with Earth's and also to upload minor course correction maneuvers and attitude realignment.

C. Simulation

An STK simulation of the communications link failed to generate usable data due to the extreme distances and novelty of the technology. STK doesn't support the optical communications settings needed, so the results would have been irrelevant. A new calculation was performed using hand calculations and tabulated into the following table.

Basic Downlink for 30m Ground Station		
Range	950	AU
Transmit Wavelength	1064	nm
Transmit Power	15	W
Transmit Gain	126.39	dB
Receive Gain	155.94	dB
Space Loss	424.5	dB
Pointing Loss	0.8	dB
Optics Loss	2	dB
Atmospheric Loss	2.5	dB
Total System Losses	429.8	dB
EIRP	138.15	dBW
Received Power	-135.71	dBW
Noise Power	-110.15	dBW
Link Margin	13.747	dB
Boltzmann's Constant	1.38E-23	J/K

Fig. 6 Downlink link budget analysis

D. Results

The STK simulation of the ground sites has shown that the three chosen ground sites can support 24/7 line of sight. The more basic link budget analysis has produced a link margin of 13.747 at 950 AU, this is a very strong value which exceeds requirements. Considerations involved in this analysis included loss during travel, pointing loss, and atmospheric losses. A more sophisticated analysis of link margin should yield a link margin of 4.43 at 900 AU. This is based on a relevant analysis done by Helvajian et al. which included atmospheric losses (wavefront error/Strehl,scintillation, absorption, Mie-scattering, high Cirrus clouds), stellar background noise, detector dark and thermal noise, and optical amplifier (LNA) noise[1].

VI. Flight Dynamics Subsystem

The Flight Dynamics Subsystem designs and analyzes the spacecraft trajectory from Earth departure to the Solar Gravitational Lens (SGL) focal region at 650 AU. The target lies on the antipodal line-of-sight to Proxima Centauri b

at RA = 37.4289°, Dec = +62.6795°. A high-energy trajectory combining a Jupiter gravity assist with solar Oberth maneuvers was selected to achieve the 650 AU distance within 30 years.

A. Requirements

The trajectory shall deliver the spacecraft to 650 AU along the target line-of-sight within 30 years. The spacecraft departs from GEO (42,164 km). The solar perihelion shall not be less than 0.046 AU (9.88 $R_{\odot}$) to limit thermal loads, and Jupiter flyby periapsis shall not be less than 1.0 R_J to limit radiation exposure. The subsystem shall provide a complete Δv budget for propulsion sizing.

B. Candidates

Four architectures were evaluated: (1) **Direct Solar Oberth** (GEO → Jupiter Gravity Assist → Solar Oberth → 650 AU)—selected as baseline for its simplicity and shortest duration; (2) **VEEJ Sequence** with Venus and Earth gravity assists—lower departure energy but longer duration; (3) **Electric Propulsion**—higher I_{sp} but >50-year mission; (4) **Solar Sail**—propellantless but immature technology.

C. Simulation

The trajectory was analyzed using a MATLAB patched-conic model that computes the Hohmann-type Earth–Jupiter transfer (2.73 years), Jupiter-centered hyperbolic flyby with powered periapsis maneuver, heliocentric fall to 0.046 AU perihelion (2.12 years), and hyperbolic escape to 650 AU (24.98 years).

Higher-fidelity analysis was attempted in AGI STK 12 Astrogator. The Mission Control Sequence included: Initial State (GEO), Target Sequence for Earth escape, heliocentric propagation to Jupiter with Sun Point Mass propagator, Target Sequence for Jupiter Oberth, propagation to solar perihelion, Target Sequence for solar Oberth, and propagation to 650 AU. Key challenges encountered include differential corrector convergence issues with the highly eccentric post-Jupiter orbit ($e = 0.98$), propagator/central-body transitions at sphere-of-influence boundaries, and constraining the outbound asymptote toward the target RA/Dec. Future work will refine initial guesses from MATLAB, increase iteration limits, and implement asymptote direction constraints.

D. Results

Figure 7 shows the STK simulation of Earth escape from GEO.

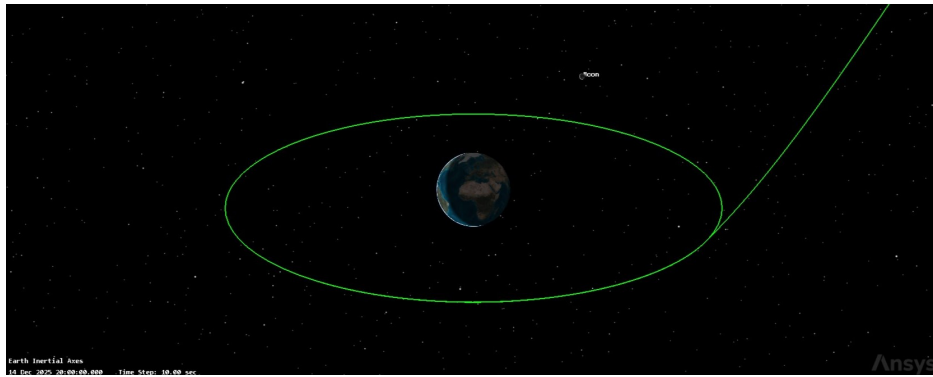


Fig. 7 STK simulation of Earth escape from GEO.

Earth Departure: From GEO, the TMI maneuver provides $\Delta v = 6.73$ km/s to achieve $v_{\infty} = 8.79$ km/s ($C_3 = 77.28$ km²/s²) for the Jupiter transfer.

Jupiter Flyby: The spacecraft arrives at Jupiter with $v_{\infty} = 5.64$ km/s and performs a powered flyby at 1.0 R_J altitude. The Oberth maneuver ($\Delta v = 1.12$ km/s) places the spacecraft on a heliocentric orbit with aphelion at Jupiter (5.2 AU) and perihelion at 0.046 AU ($e = 0.9825$).

Solar Oberth: At perihelion, the spacecraft velocity reaches 195.53 km/s (local escape velocity: 196.39 km/s). The solar Oberth maneuver ($\Delta v = 36.34$ km/s) achieves escape at $v_{\infty} = 123.25$ km/s (26.0 AU/year), reaching 650 AU in 24.98 years. The Oberth effect saves 86.9 km/s compared to a deep-space burn.

Table 2 Mission Δv budget and timeline. (Appendix XII.A)

Maneuver	ΔV (km/s)	%	Cumulative Time
Trans-Jupiter Injection (GEO)	6.73	15.2	0 yr
Jupiter Oberth	1.12	2.5	2.73 yr
Solar Oberth (0.046 AU)	36.34	82.2	4.85 yr
Total	44.19	100	29.83 yr to 650 AU

Propulsion Feasibility: The 44.19 km/s total exceeds chemical propulsion capability. With $I_{sp} = 450$ s, the required mass ratio is $m_0/m_f \approx 22,000$, which is infeasible. Alternative propulsion (nuclear thermal, solar thermal, or solar sails) will be evaluated in future trades. The trajectory geometry established here remains valid regardless of propulsion selection.

All flight dynamics requirements are satisfied: 650 AU reached in 29.83 years along the target line-of-sight, with perihelion and flyby altitude constraints met.

VII. Propulsions Subsystem

A. Requirements

Propulsion serves an important role in this long distance mission. The goal of the analysis is to provide an effective means of satisfying the ΔV required at each point in the time allowed. These points include the initial burn from GEO, the flyby burns at Jupiter and the Sun, and station keeping once in deep space. Equally as important is trying to keep the mass at a minimum using current or in development propulsive technology. Additionally, the propulsion system must be able to handle any environmental factors that occur during the mission.

B. Candidates

1. Chemical Propulsion

This method is extremely useful in situations that require a high thrust, such as the oberth maneuvers done at the Sun and Jupiter. However, bi-propellant has a relatively low specific impulse, with the best current methods of non-cryogenic fuel being NTO/N₂H₄ (Nitrogen Tetraoxide and Hydrazine) providing an I_{sp} of 338s.[16]

2. Ion Propulsion

Ion propulsion is a potential solution that provides a large amount of ΔV with relatively low needed fuel mass along with incredible minimum impulse. However, it comes with a trade off of requiring lots of power and providing very little thrust. However, in the deep space part of the mission the thrust required is very little and so mini ion thrusters can be used for their precise control.

3. Solar Sail

This is the most promising of the bunch for achieving high ΔV for low mass. However, it is limited by current technology in materials and manufacturing. [17] More details about its performance and size will be shown in the results.

C. Simulation

1. Sun Oberth Burn

The most useful way to think about this mission is to start from the end and work backwards from there. Most of the "simulation" here is using theoretical calculations as they are the quickest method to determine if something is viable. More precise calculations will be done with computer simulations in the future.

In order to delve into this, some math must be developed using the solar sail. The force of a solar sail is given by:

$$F_s = F_0 A \cos^2(\theta) (1 + \rho) \left(\frac{1 \text{ AU}}{r} \right)^2 \quad (8)$$

where F_0 is $4.54 \frac{\mu\text{N}}{\text{m}^2}$ at 1 AU, A is the area of the sail, θ is the angle between the normal vector of the sail with the position vector relative to the Sun, ρ is the reflectivity of the sail, and r is the distance from the center of the Sun. Because the force scales with the inverse square, we can actually convert this into an effective gravitational parameter:

$$\mu_{eff} = \mu_s - F_0 \frac{A}{m} (1 + \rho) (1 \text{ AU})^2 = \mu_s - k_s R \quad (9)$$

where μ_s is the gravitational parameter of the Sun and m is the mass of the spacecraft. In our analysis, the mass will include the station keeping fuel and the mass of the solar sail, but not the chemical fuel that will be used in the solar burn. Note the simplification made here, where R is the sail area to spacecraft ratio, as that will become important later.

Another characteristic of the sail is its areal density ρ_A . Using this we can find the area to mass ratio of the spacecraft without the sail, which is much more useful in our analysis.

$$R_0 = \frac{R}{1 - \rho_A R} \quad (10)$$

In order to determine the amount of fuel needed to burn, we can use conservation energy to get the required ΔV from just the fuel:

$$\Delta V = \sqrt{V_\infty^2 + \frac{2\mu_{eff}}{r_p}} - V_p \quad (11)$$

where r_p is the radius of perihelion at 0.14 AU with the sail and V_p is velocity at perihelion which from the last section is 114 km/s. Using the rocket equation we get:

$$\frac{m_3}{m_4} = \left(\frac{1}{1 - \rho_A R} \right) e^{\frac{\Delta V}{I_{sp} g_0}} \quad (12)$$

Where m_4 is the mass of the craft only including dry mass while m_3 also includes the fuel mass required for the Sun Oberth and the mass of the solar sail. Our goal is to find the sail area required to minimize that ratio. When taking the derivative with respect to R and finding the value for the minimum, we get:

$$R_{min} = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \quad (13)$$

where:

$$a = r_p \rho_A^2 \quad (14)$$

$$b = 2S k_s - 2r_p \rho_A \quad (15)$$

$$c = r_p - S r_p V_\infty^2 - 2S \mu_s \quad (16)$$

$$S = \left(\frac{\rho_A r_p I_{sp} g_0}{k_s} \right)^2 \quad (17)$$

plugging this back into mass ratio equation will give the minimum mass ratio.

2. Jupiter Transfer and Oberth Burns

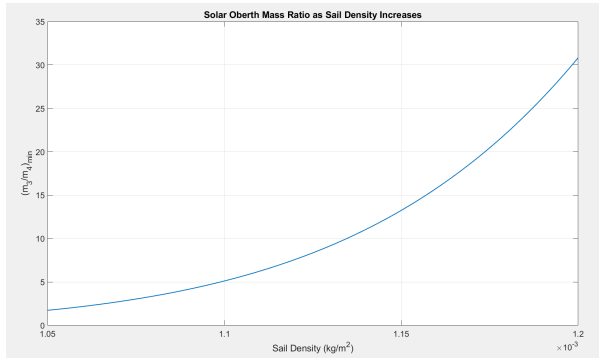
This section is much simpler. To solve this, a rocket equation is all that is needed. We get that for a both burns, using an I_{sp} of 338:

$$\frac{m_2}{m_3} = 1.40, \frac{m_1}{m_2} = 7.61 \quad (18)$$

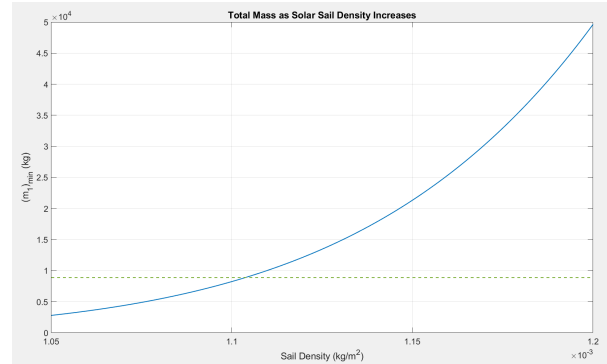
where m_2 is the mass of the craft right before Jupiter Oberth burn and m_1 is the net weight of the craft, before Trans-Jupiter Injection.

D. Results

The variables that have the most effect on the mass of the aircraft is the total fuel needed to do the first two transfers, which is limited by propulsive technology, but also the areal density of solar sail.



(a) This graph depicts the mass ratio as sail density increases. Note that the 14% increase in areal density increases the mass by 30x.



(b) This graph depicts the final mass of the spacecraft along with the highest mass allowed by Atlas V for their GEO orbits. The highest allowable sail density occurs at 1.104 g/m²

The graphs indicate that an areal density of 1.104 g/m² is necessary in order for the payload mass, which is 8900 kg, to be light enough to fit on the Atlas V rocket.[18] The solar sail area at this density is 5.9 million m², which is quite significant. Volume wise, assuming a thickness of 1 micron, the sail will take up 5.9 m³ of space.

The deployment of a sail of this size is unrealistic however and there are many factors that have not been considered like the fuel needed for station keeping once out into deep space. Other options, such as ion propulsion, have been looked at. However, they are difficult to use as the power available is quite low for these missions. Additionally, simulations of solar sails would be required after analysis deems them usable in order to get more predictable outcomes. This is for future research and is planned to be at least partially completed next year. Lastly, once a set of propulsion systems and their fuel have been determined, piping and containment of that fuel can begin.

VIII. Attitude Determination and Controls Subsystem

A. Overview

The main purpose of this subsystem is to maximize our chances of success in cases where we may not be fully stable or in control. Understanding the attitude of your system is crucial to minimizing the chances of errors and inaccurate output when in deep space. That is why we will be using tools such as multiple reference and Inertial sensors, a star tracker, and reaction wheels. These tools will allow us to determine the attitude of our satellite as well as to give us immediate feedback on flight control information for other subsystems that come into play. Our goal is to make these tools have different modes of use. Passive vs active. That is, we'll want to manually implement our data before going into fully autonomous control of these tools, as they will be in deep space for a long period of time. The selection of these devices was made based on high performance and accuracy. The research made on these devices provided us with the proper results and feedback to make our final decision.

B. Requirements

For our satellite we will be implementing a camera on our star tracker. This will allow us to get the most accurate photos of anything we seek to capture in deep space. In order to do this, first we must define our camera itself with a FOV of 8-20 degrees. In the design process we need to consider exposure and dynamics, including dynamic range to avoid bright-star saturation. This will allow us to avoid motion blur on all images taken. Next is Adjusting attitude, we will need to use the TRIAD estimator method once we find our body-frame vectors that come from the star's centroids.

When it comes to our Solar Sail we will want to implement some hardware on there as well. We know our solar sail will eventually need to be perpendicular to the sun for maximum propulsion. That is why we will be adding 4 sun sensors to our sail. We will place 3 on the face of the sail and 1 on the back where it's shadowed by the sail for position

feedback (4 total). These types of sensors will all be digital. This specific sensor placement will capture full coverage of the sail when pointed properly. In this case, we can allow for natural movements in the aircraft until it gets to 650 AU, as long as it is working towards turning the our telescope around to point at the sun. Once we reach 650 AU, all propulsion is done and our solar sail will need to flip around so that our star tracker camera can be pointed properly. This is why implementing a star sensor onto the camera will allow us to acquire proper feedback flight control from both sensors used. We will be having to turn one on and off simultaneously so our readings aren't conflicted at high speeds.

C. Simulation

When constructing our design we sought to have the most accurate simulation design we could implement. That is why utilizing both reference and inertial sensors in each component would allow us to get the best results. When it comes to Reference sensors, these measure the direction to an object and provide a definite “fix”. In general we know the accuracy reference sensors increases with distance to the object. However, in some cases these are not always available to use. For example a sun sensor is useless during a solar eclipse. That is why we must also implement Inertial sensors. These will measure continuously and they measure “changes” in the attitude of our system. Moving forward they will need a “fix” which is a calibration that comes from reference sensors readings.

Stars in space are usually fixed so utilizing the star tracker we may determine its relative position to our satellite. In our case we need to use multiple sensors that can scan or detect multiple objects (like planets or stars) in a short time frame. This is because at some point we will be moving at 650 AU.

For our Star Tracker, we will be implementing software that allows for a 3-axis attitude as well as being able to follow the motions of the satellite itself with embedded star catalogs and constellations. We will have a Narrow and wide performance Field of view. These will both weight about 2.5 – 3 kg but essentially the same power output of 10 Watts with an input of 10 Volts. In addition we will be implementing reaction wheels. Our reaction wheel will serve us in a 4-wheel pyramid formation evenly spaced small-sat wheels. Each equipped with 18-20 Watts of power. This means no thrusters will be required for our wheels and in order to control these we will use a fully autonomous Attitude Control System Design (ACSD) computer with a gyro and most importantly our star tracker's input when in motion.

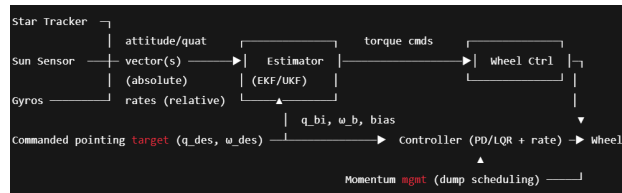


Fig. 9 ADCS Control Diagram

D. Results

The Attitude Estimates that we are taking into consideration for our satellite are mainly the accuracy and requirements when pointing our telescope. Specifically, for these Pointing requirements, we may be able to improve our pointing accuracy, stability, and jitter control. How can this be done? Incorporating the next generation of sensors like an Ultra-high precision star tracker whose performance trends shift from 1-10 arcsec down to 0.1-1 arcsec. These types of sensors are now fused with AI and provide multi-FOV on the camera itself. When we look deeper into our star tracker, when adjusting the Camera's attitude we will need to include filtering and fusion. Simply put when our tracker is in “tracking mode” we need to combine the tracker attitudes and gyro rates to prevent things like dropouts, jitter, which will allow for shorter exposure. When calibrating we need to focus on reference points in orbit, these can be found using AI-Enhanced attitude determination tool to find star centroids. Finally, when it comes to specific handling and harsh environments, the pointing accuracy of our telescope may be affected by thermal effects. We need to take into consideration the effects of radiation and high slew (the rapid change in outputs) which can cause inaccurate imaging from our camera. For example, there are cases where pixels can rise to ridiculously high temperatures and it can end up creating “false stars” in the camera's imaging. This is caused by the radiation damage. However, these cases can be mitigated with proper masking or shielding on our part.

IX. Mechanical Subsystem

A. Overview

The Mechanical Subsystem provides the primary structural framework of the spacecraft, supporting all major components during launch, the solar Oberth maneuver, and the extended deep-space cruise. It consists of a stiff cylindrical truss that maintain geometric stability and prevent buckling under the launch load of the spacecraft. This structure ensures proper alignment of the telescope payload, secure mounting of propulsion and tank systems, and interfaces for the solar sail and radiation-shield deployments.

B. Requirements

The Mechanical Subsystem provides the primary load-bearing framework for the spacecraft and must support all other subsystems during launch, solar sail deployment, and the long-duration cruise to the Solar Gravitational Lens focal region. The structure must maintain the alignment of the 1 m primary mirror and deployable 0.25 m secondary mirror at the forward end, support the propellant tanks and propulsion hardware near the vehicle's center, and provide mounting for the sail deployment mechanisms and radiation umbrella at the aft end. During launch, the frame is required to withstand a 10 g axial acceleration acting on a total spacecraft mass of 1356 kg, which corresponds to an axial compressive load of approximately 1.33×10^5 N. This load must be carried without exceeding the yield strength of Aluminum 6061-T6 and without inducing buckling in any of the main structural members. In addition, the structure must provide sufficient bending stiffness to avoid excessive deflections and to achieve an acceptable first bending-mode frequency for compatibility with launch vehicle dynamic requirements, while remaining manufacturable and mass-efficient for a long-duration deep-space mission.

C. Candidates

Several structural configurations were considered to satisfy these requirements. Early concepts included a thin-walled cylindrical shell and a multi-longeron truss with smaller members, but these either produced insufficient buckling margins or were less practical to fabricate. The selected configuration is a four-longeron cylindrical truss constructed from Aluminum 6061-T6 C-channels with a nominal depth of 60 mm, flange width of 40 mm, and wall thickness of 8 mm. Five circular rings connect the longerons: two rings at the ends of the 8 m frame and three interior rings spaced at 2 m intervals. Each ring incorporates an internal X-brace, which ties the longerons together and reduces the effective buckling length of each member to the 2 m spacing between rings. This design distributes axial and bending loads efficiently, provides clear attachment points for the payload, tanks, and thrusters, and offers geometric stability over the mission lifetime. The design is visually demonstrated in Figure 10 below

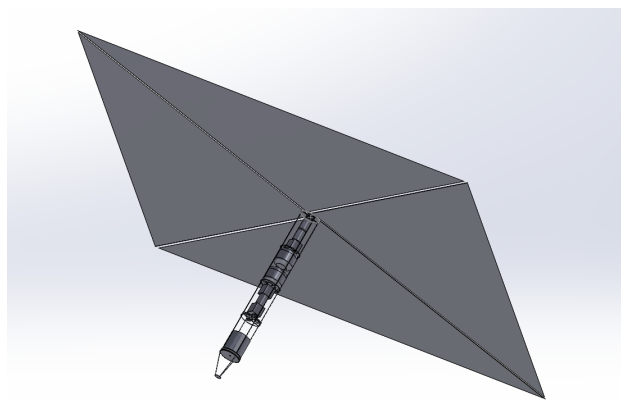


Fig. 10 3D Model of Structural design

Aluminum 6061-T6 was chosen because it is widely available, easy to machine, and compatible with conventional joining methods such as bolting and TIG welding. The C-channel profiles can be obtained from commercial suppliers as extrusions or machined from rectangular bar, and the rings and X-braces can be cut from plate or strip using waterjet or CNC processes. Joint details can be implemented as welded gussets or bolted clevises at the ring-longeron intersections,

allowing the frame to be built with modest manufacturing resources while achieving the dimensional accuracy needed for optical alignment. This combination of a four-longeron deep C-channel truss with X-braced rings was therefore selected as the baseline mechanical architecture.

D. Simulation

A first-order structural analysis of the selected truss configuration was carried out in MATLAB. The launch condition on the spacecraft yields a total axial compressive load of $F_t = 1.3310^5$ N, which is assumed to be shared equally among the four longerons, giving $F_l = 3.3310^4$ N per member. For the chosen $60 \times 40 \times 8$ mm C-channel, the cross-sectional area is approximately $0.00112m^2$, producing an axial stress of about 30 MPa, well below the 276 MPa yield strength of Aluminum 6061-T6 and giving a yield safety factor of roughly 9.3.

Buckling was evaluated using Euler's column formula for a pinned-pinned segment with effective length $L_{eff} = 2$ m, corresponding to the spacing between rings. The second moment of area of the C-channel about its strong axis was computed using an approximate "solid rectangle minus inner cutout" representation. With these properties and the aluminum modulus $E = 68.9$ GPa, the critical buckling load for each 2 m segment was found to be approximately 8.410^4 N. Comparing this with the applied axial load per longeron gives a buckling safety factor of about 2.5.

The solar sail booms were analyzed separately using a similar Euler buckling approach. For 7 m long aluminum booms with 20 mm outer diameter and 0.3 mm wall thickness, the compressive load induced by solar radiation pressure at 1 AU is only about 2.310^4 N per boom. An equivalent-beam model of the overall truss, treating the four longerons at a 0.5 m radius and including their parallel-axis contributions, was used to estimate the first bending-mode frequency. Using the full spacecraft mass of 1356 kg and the computed equivalent bending inertia, the fundamental bending frequency was found to be approximately 2.1 Hz.

The MATLAB script also performed parametric sweeps over wall thickness and section depth, generating plots of axial stress versus thickness and buckling safety factor versus thickness and depth. These parametric results confirm that increasing both the depth and wall thickness of the C-channel significantly improves buckling performance compared to earlier, shallower designs. These results are displayed in Figure 11.

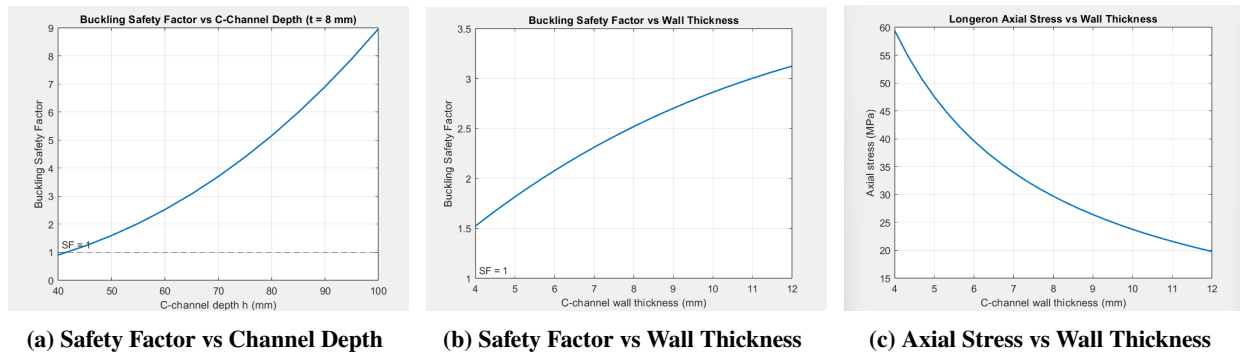


Fig. 11 Structural response metrics as a function of channel depth and wall thickness

E. Results

The updated analysis shows that the four-longeron truss built from $60 \times 40 \times 8$ mm Aluminum 6061-T6 C-channels provides adequate strength and stability for the 10 g launch condition of the 1356 kg spacecraft. The axial stress under full launch loading remains well within allowable limits, with a yield safety factor greater than 9, and the Euler buckling analysis for the 2 m braced segments yields a safety factor of about 2.5, indicating that column buckling is not a limiting failure mode for the selected design. The ring and X-brace system effectively constrains the longerons laterally and ties the frame into a stiff, self-supporting structure. The solar sail booms are far from their buckling limits under nominal radiation pressure loads and do not govern the mechanical design. The estimated first bending-mode frequency of roughly 2.1 Hz is lower than typical launch vehicle preferences, suggesting that global stiffness could be improved further through additional bracing, increased truss diameter, or refined joint design in future iterations. More detailed and accurate analysis will be required as the 3D model is fully developed. Weight distribution had to be emulated by considering each subsection as a uniformly dense block, which will affect the moment of inertia calculations, and Finite Element Analysis will give a more detailed look at the stresses experienced once the design is finalized.

X. Thermal Subsystem

The thermal subsystem is responsible for maintaining the spacecraft bus and its critical components within an allowable temperature range throughout all phases of the mission. This includes extreme conditions near the Sun when performing our solar Oberth maneuver to surviving in deep space for decades on our cruise beyond the solar system to the focal region. The thermal subsystem is designed to prioritize thermal regulation mechanisms, use quality surface thermo-optical properties, ensure proper insulation, and to use nuclear heat sources to try and minimize complexity and overall power consumption. A simulation was run using a first order radiative equilibrium analysis to evaluate our thermal performance, to establish quantitative design requirements, and to verify the compliance of our mission constraints. The following sections will present the thermal requirements, technology candidates, the methodologies from our simulation, and performance assessments from our results.

A. Requirements

The thermal subsystem needs to maintain all of the spacecraft bus components and its avionics within an allowable temperature range throughout all of our mission phases. This includes our Oberth maneuver near the Sun, and when we are far out into a deep space cruise to the focal lens region at 650 AU. Requirements for the thermal subsystem will be determined on how well our spacecraft survives, the operating limits of the on board avionics, and the power availability we have long term. The spacecraft bus will need to remain within an allowable temperature range of $T_{\min} = 273$ K for our minimum temperature and $T_{\max} = 323$ K for our maximum temperature. We selected these bounds so that we can ensure we have a reliable operation of the onboard electronics while also minimizing the amount of thermal damage and material degradation we get on the spacecraft overtime. We also need to limit the absorbed solar flux that the spacecraft gets during our solar Oberth maneuver at a minimum heliocentric distance of 0.046 AU. As a result, this requirement will pose constraints on the geometry of our heat shield, the surface thermo-optical properties, and our pointing accuracy while performing our Oberth maneuver. In the deep-space cruise beyond 100 AU where solar heating from the Sun becomes negligible, we will need to prevent overcooling for the duration of our mission that will last at least 25 years. Therefore, the survival of our spacecraft will largely depend passive insulation and a continuous use of nuclear sources.

B. Candidates

We have evaluated several different technology candidates for our thermal control to try and address the extreme thermal environments we will be encountering throughout the duration of the mission. We based some of these candidates off of their thermal performance, mass efficiency, success from past missions, and compatibility with the propulsion and structural configurations of our spacecraft. When the spacecraft is very close to the Sun for performing the solar Oberth maneuver, we will make use of a carbon-carbon (C/C) composite heat shield with a reflective ceramic coating as our primary source of thermal protection. This material was chosen because it offers a high temperature tolerance, a low coefficient of thermal expansion, and proven success from missions like the Parker Solar Probe. The applied reflective coating will reduce solar absorptance from the Sun, and the carbon-carbon structure will give us robust thermal protection. Once we have finished our Oberth maneuver away from the Sun, the deployed solar sail will make use of aluminized colorless polyimide (CP1) since it provides us with a low absorptance and a high reflectivity. The idea behind this is that it will reduce the amount of thermal loading we receive while giving us more thrust from solar radiation pressure. The backside of the sail will also be designed to emit and reject heat that is internally generated. When we are far out into our deep space cruise, the primary candidate of cold protection will be multilayer insulation (MLI). The MLI blankets will be covering us most of the bus exterior, which will reduce the emissivity and preserve the internal heat generated by the electrical components. Thermal equilibrium will also be achieved by using radiators with a high infrared emissivity to reject an excess amount of internal heat. Radioisotope heater units (RHUs) and waste heat from the radioisotope thermoelectric generator (RTG) will also be considered to provide us with a continuous source of non-electrical heating. These nuclear heat sources are very important because they will provide us with decades of thermal support along with a reduced reliance of powered survival heaters that require electricity. Electrical trim heaters will only be used in parts of the spacecraft that do not have the best thermal coverage.

C. Simulation

The thermal performance of our spacecraft was evaluated using a first order steady-state equilibrium model. Since heat transfer on the spacecraft is primary dominated by radiation beyond low Earth orbit, we modeled the spacecraft as a gray body that exchanges energy within the surrounding space environment. The thermal balance for the spacecraft is

expressed as

$$\alpha J(r) A_{\text{proj}} + Q_{\text{int}} = \sigma T^4 (\epsilon_{\text{MLI}} A_{\text{MLI}} + \epsilon_{\text{rad}} A_{\text{rad}}) \quad (19)$$

where $J(r) = J_{1 \text{ AU}}/r^2$ represents the solar irradiance as a function of heliocentric distance. The equilibrium temperature T is obtained by enforcing a balance between the absorbed solar radiation, internally generated heat, and infrared emission going into deep space. The hot and cold case configurations were separated so that we can determine the thermal behavior of the craft across all mission phases. For the hot case of being near the Sun, the spacecraft bus was assumed to be protected behind the shadow of the heat shield during the solar Oberth maneuver. For the cold case being out in deep space, we assumed that the radiator emissivity was restricted and that the multilayer insulation covered most of the bus exterior to try and minimize radiative heat loss. We also modeled the internal heat dissipation as being constant throughout the duration of the cruise. This simplified model was implemented into MATLAB to evaluate the equilibrium temperatures of the spacecraft at certain heliocentric distances from perihelion to 650 AU. The purpose of this simulation is to evaluate whether or not the spacecraft can stay within the survivable temperature range until it reaches the focal region.

D. Results

After applying the first order radiative equilibrium model to the spacecraft, we can safely confirm that the thermal architecture will satisfy our mission level requirements across all phases of the mission. For the hot case of being near the Sun for the solar Oberth maneuver, the analysis indicates that the spacecraft bus will be maintained below the maximum allowable limit within the shadow of the heat shield cone. The math therefore tells us how much sunlight is permitted to leak past the heat shield, how big the heat shield needs to be, and accurately the spacecraft must point at the Sun during its Oberth maneuver. For the cold case out in deep space at around 100 AU, this model predicts an equilibrium temperature of roughly 278 K. This result exceeds the minimum allowable temperature and demonstrates that the spacecraft can survive without continuous electrical heating. It is important to note that small trim heaters may still need to be used for components that are thermally isolated. More detailed thermal modeling will be performed in the following semester once our spacecraft geometry and subsystem interfaces are finalized.

XI. Conclusion

The goal of this mission is to achieve the dream that many scientists have: a picture of a solar system outside ours. Not only will these spur many advancements in a plethora of fields, but also it will help confirm or deny the many theories about the formation of planets, stars, and the universe. The mission also has a exciting secondary objective of possibly confirming the existence of life besides the ones on this planet.

However, a mission of this size requires many advanced technologies to support it. Innovations in data processing are required to be able to fit within power requirements. Additionally, an increase in the reliability of energy storage systems are needed to keep the spacecraft surviving in the cold of space. Along with that, deep research will have to be done into optimizing communications from far away in order to support a large data rate while using little power. In order to make sure the spacecraft is not unreasonably massive, advancements in simulation and especially materials for solar sails are necessary. However, simulations can not completely predict everything, so position determination is absolutely crucial. With such a large distance, new advancements must be made in star tracking accuracy to allow such precise positioning of the spacecraft in order to achieve high quality images of exoplanets. Lastly, to support the frigid temperatures and excessive radiation in deep space, a deep analysis of data from previous deep space missions must be utilized to optimize better materials for environmental protection of the craft.

Although this task may seem daunting, the benefits it will bring to the pursuit of science and technology is greater than the difficulty. With the advancements in communication, satellites can go deeper into space than even the Voyager craft did. Additionally, the significant advancements in materials can reduce weight, and therefore fuel, on spacecrafts in the future, reducing the amount of limited resources humanity will use in the long term. Moreover, innovations in propulsion will allow inter-planetary travel to become much cheaper and available, creating a pathway for humans to rise from Earth and conquer the solar system. It's not a stretch to say that this project, if completed, can change the scientific landscape as we know it.

XII. Appendix

Nomenclature

A	=	Surface area
$ADCS$	=	Attitude Determination & Control Subsystem
AU	=	Astronomical Unit
CAD	=	Computer-Aided Design
$COTS$	=	Commercial Off-the-Shelf
DC	=	Direct Current
DOD	=	Depth of Discharge
DSN	=	Deep Space Network
$EDFA$	=	Erbium-Doped Fiber Amplifier
EPS	=	Electrical Power Subsystem
GRC	=	Glenn Research Center
$JWST$	=	James Webb Space Telescope
$J(r)$	=	Solar irradiance at heliocentric distance
$J_{1\text{ AU}}$	=	Solar constant at 1 AU
$LCRD$	=	Laser Communications Relay Demonstration
LEO	=	Low Earth Orbit
LNA	=	Low-Noise Amplifier
m_1	=	Net Mass of Spacecraft
MHW	=	Multi-Hundred Watt
$MHW-RTG$	=	Multi-Hundred Watt Radioisotope Thermoelectric Generator
$MOPA$	=	Master-Oscillator Power Amplifier
$NASA$	=	National Aeronautics and Space Administration
NiH_2	=	Nickel-Hydrogen Battery
OBC	=	On-Board Computer
PCb	=	Proxima Centauri b
PCU	=	Power Conditioning Unit
PPM	=	Pulse Position Modulation
Pu	=	Plutonium
Q_{int}	=	Internal heat dissipation
r	=	Heliocentric distance (AU)
$RealTOR$	=	Real Time Optical Receiver
RF	=	Radio Frequency
RPS	=	Radioisotope Power System
RTG	=	Radioisotope Thermoelectric Generator
S/C	=	Spacecraft
SGL	=	Solar Gravitational Lens
SNR	=	Signal to Noise Ratio
STK	=	Systems Tool Kit
T	=	Spacecraft bus equilibrium temperature
$TT\&C$	=	Telemetry, Tracking, & Control
V	=	Volt
W	=	Watt
α	=	Effective solar absorptance of the spacecraft bus exterior
ϵ	=	Emissivity
μ_{eff}	=	Effective Solar Gravitational Parameter
ρ	=	Reflectivity
ρ_A	=	Sail Areal Density
σ	=	Stefan–Boltzmann constant

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A. Code

Payload

Data Handling

Electrical

TT&C

Flight Dynamics

`https://github.com/wyattowelch/AE460-Solar-Gravitational-Lens/blob/main/SGL\_Mission\_FlightDynamics.m`

Propulsions

`https://github.com/wyattowelch/AE460-Solar-Gravitational-Lens/blob/main/FinalSailCode.m`

ADCS

Mechanical

`https://github.com/wyattowelch/AE460-Solar-Gravitational-Lens/blob/main/StructureStressAnalysis.m`

Thermal

References

- [1] Helvajian, H., Riley, R., Sercel, J., Etchegaray, J., Landis, G., and Garrison, J., “A Mission Architecture to Reach and Operate at the Focal Region of the Solar Gravitational Lens,” *Journal of Spacecraft and Rockets*, Vol. 49, No. 1, 2012, pp. 118–129. <https://doi.org/10.2514/1.A35493>, <https://arc.aiaa.org/doi/epdf/10.2514/1.A35493>.
- [2] NOIRLab, “The Charge-Coupled Device: Revolutionizing How Astronomers See the Universe After 50 Years of Development, What’s on the Horizon?” <https://noirlab.edu/public/blog/50-years-ccds/>, 2024. NOIRLab Blog, 22 April 2024.
- [3] Kihm, H., Yang, H.-S., and Lee, Y.-W., “Optomechanical Analysis of a 1-m Light-weight Mirror System,” *Journal of the Korean Physical Society*, Vol. 62, No. 9, 2013, pp. 1239–1246. <https://doi.org/10.3938/jkps.62.1239>, URL <https://link.springer.com/article/10.3938/jkps.62.1239>.
- [4] Turyshev, S. G., and Toth, V. T., “Resolved imaging of exoplanets with the solar gravitational lens,” *arXiv preprint*, 2022. <https://doi.org/10.48550/arXiv.2204.04866>, also published as MNRAS 515 (4), 6122–6132 (2022).
- [5] Zhou, H., “SGL Coronagraph Simulation,” Presentation at the Keck Institute for Space Studies workshop “Technology Requirements to Operate at and Utilize the Solar Gravity Lens for Exoplanet Imaging”, 2018. URL <https://www.kiss.caltech.edu/workshops/solar/presentations/Zhou.pdf>, pasadena, CA, May 15–18, 2018.
- [6] Raj, R., and Billa, K., “Post-launch vicarious calibration-validation of Pixxel’s technology demonstrator VNIR hyperspectral satellite using RadCalNet data,” https://www.researchgate.net/publication/397978403_Post-launch_vicarious_calibration-validation_of_Pixxel%27s_technology_demonstrator_vnir_hyperspectral_satellite_using_RadCalNet_data, 2025. Preprint, available on ResearchGate.
- [7] Toth, V. T., and Turyshev, S. G., “Image recovery with the solar gravitational lens,” *arXiv preprint*, 2020. <https://doi.org/10.48550/arXiv.2012.05477>, also published as Phys. Rev. D 103, 124038 (2021).
- [8] Welch, Wyatt O., “AE460 Solar Gravitational Lens Mission Simulation and Data Handling Pipeline,” , 2025. URL <https://github.com/wyattowelch/AE460-Solar-Gravitational-Lens>.
- [9] NASA, “Radioisotope Power Systems Overview,” NASA Science Mission Directorate, 2023. URL <https://science.nasa.gov/planetary-science/programs/radioisotope-power-systems/overview/>, accessed via NASA Science.
- [10] Schock, A., and Or, C., “Radioisotope Thermoelectric Generator Thermal Management,” Tech. Rep. NASA-TM-2005-213431, NASA Jet Propulsion Laboratory, 2005. URL <https://ntrs.nasa.gov/api/citations/20050238992/downloads/20050238992.pdf>.
- [11] NASA, “Electrical Power System Standardization for Spacecraft,” Tech. Rep. NASA-SP-100, NASA, 1985. URL <https://ntrs.nasa.gov/citations/19850005601>.
- [12] Manzo, M. A., “Nickel-Hydrogen Battery Handbook,” Tech. Rep. NASA-HDBK-4001, NASA Goddard Space Flight Center, 1995. URL <https://ntrs.nasa.gov/api/citations/19950010446/downloads/19950010446.pdf>.
- [13] Bennett, G. L., “Multi-Hundred Watt Radioisotope Thermoelectric Generator (MHW-RTG) Performance,” Tech. Rep. NASA-TM-2016-219072, NASA, 2016. URL <https://ntrs.nasa.gov/api/citations/20160001769/downloads/20160001769.pdf>.
- [14] Hemmati, H., Mandl, D., Wilson, K., Lambie, C., Grunes, M., and Israel, D., “Optical Communications for Interstellar Precursor Missions,” Tech. Rep. NASA/TM–20240012961, NASA Jet Propulsion Laboratory, March 2024. <https://ntrs.nasa.gov/api/citations/20240012961/downloads/TM-20240012961.pdf>.
- [15] Segal, J., Gofman, A., and Castiel, A., “Laser Communications at 1000 AU: Concept and Technologies,” Tech. Rep. ADA564661, Defense Technical Information Center, 2012. <https://apps.dtic.mil/sti/pdfs/ADA564661.pdf>.
- [16] Reed, B. D., “On-Board Chemical Propulsion Technology,” Tech. Rep. 20040061966, NASA, 2004. URL <https://ntrs.nasa.gov/api/citations/20040061966/downloads/20040061966.pdf>.
- [17] Christensen, C., “Ultralight Solar Sails for Interstellar Travel: Phase I Final Report,” Phase I Final Report NIAC CP 99-02, NASA Institute for Advanced Concepts (NIAC), Atlanta, GA, Dec. 1999. URL https://www.niac.usra.edu/files/studies/final_report/333Christensen.pdf.
- [18] United Launch Alliance, “Atlas V,” , 2025. URL <https://www.ulalaunch.com/rockets/atlas-v>, accessed December 15, 2025.